



Review Article

Advances in Anesthesia Techniques for Postoperative Pain Management in Minimally Invasive Cardiac Surgery: An Expert Opinion

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Minimally invasive cardiac surgery (MICS) often leads to severe postoperative pain. At present, multimodal analgesia schemes for MICS have attracted much attention, and the application of various chest wall analgesia techniques is becoming increasingly widespread. However, research on anesthesia techniques for postoperative pain management in MICS remains relatively limited at present. We searched for relevant literature and summarized recent related research in eight MICS techniques, including thoracic epidural anesthesia, spinal anesthesia, thoracic paravertebral plane block, erector spinae plane block, serratus anterior plane block, pectoral nerve block, intercostal nerve block, and parasternal block. This article provides an overview of the anatomy and procedures involved in these analgesic techniques, their mechanisms of action, and the latest clinical trial evidence. It also evaluates their progress in MICS, compares their advantages and disadvantages, and discusses practical challenges.

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PAIN AFTER THORACIC and cardiac surgery typically originates from multiple sources, including surgical incisions, rib and sternum injuries, damage to intercostal nerves, trauma to visceral tissues, and irritation or tension from thoracic drainage tubes. These pain sources involve distinct pathophysiological mechanisms. Somatic pain includes discomfort from the incision itself, rib and intercostal nerve damage, and pain associated with chest drainage tube placement. This type of pain can range from mild to severe, due to bone structure changes and nerve damage. It is typically localized and can be exacerbated by patient movement, often leading to poor sleep, anxiety, and hindering postoperative rehabilitation. In contrast, visceral pain is primarily transmitted through the sympathetic nervous system. The trachea and esophagus, similar to the surgical site, are primarily innervated by the parasympathetic nervous system. Visceral pain tends to be slow, persistent, and diffusely located. Both types of pain can cause significant

discomfort for patients.¹ Postoperative pain not only increases health care costs, but also negatively impacts quality of life. It affects the patient's ability to resume normal activities, extends the recovery period, and can impair respiratory function, potentially leading to pulmonary complications.¹ Therefore, effective postoperative pain management is crucial to minimize atelectasis and cardiovascular issues, and to facilitate earlier extubation, mobilization, and discharge.²

Studies indicate that patients undergoing sternotomy for cardiac surgery frequently experience mild to moderate postoperative pain, which may progress into chronic discomfort. Specifically, 20% of patients report persistent mild pain, and 3% continue to experience moderate pain 3 months after surgery.³ In contrast with traditional sternotomy, minimally invasive cardiac surgery (MICS) is increasingly favored due to its advantages, such as smaller incisions, reduced tissue trauma, less bleeding, and shorter hospital stays.⁴

Minimally invasive surgical techniques are now widely used across various cardiac procedures, including coronary artery bypass grafting, aortic valve replacement, and mitral valve replacement. Additionally, robotic-assisted procedures, such as atrial myxoma excision and septal myectomy, have become

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increasingly prevalent.⁵ These surgeries require distinct approaches and incisions. The most commonly used method for aortic valve surgery is hemisternotomy, involving a mid-line incision on the upper one-half of the sternum, extended 4 to 5 cm inferiorly. Alternatively, the right anterior thoracotomy offers a minimally invasive technique for aortic valve surgery, characterized by a 4- to 6-cm incision located at the third intercostal space on the right side. In the context of mitral valve interventions, the right thoracotomy has emerged as the predominant minimally invasive approach, characterized by a 4- to 6-cm incision positioned along the midaxillary line at the fourth intercostal space. Robotic surgery employs a similar incision to right thoracotomy, but with two 10-mm incisions through which robotic arms are inserted into the chest cavity. Totally endoscopic coronary artery bypass surgery typically involves harvesting the internal mammary artery with a robotic arm, followed by vascular anastomosis through a 6- to 10-cm incision in the fifth intercostal space. For direct coronary artery bypass surgery, a 4- to 7-cm anterior lateral thoracotomy is often performed in the fifth intercostal space.⁶

It has been established that different surgical techniques have distinct impacts on postoperative pain. Seven days after surgery, the pain scores for procedures involving rib cartilage resection were the highest, whereas the lowest pain scores were observed in the group undergoing total endoscopic cardiac surgery. In a comparable manner, individuals who experienced rib cartilage resection exhibited the greatest intake of analgesics within the 30-day postoperative period, whereas those who underwent total endoscopic cardiac surgery demonstrated the lowest requirement for analgesic use.⁷ Although minimally invasive surgeries typically involve smaller incisions and result in less postoperative pain than traditional methods, effective pain management remains a critical concern.⁸ An Italian study reported that, among patients who underwent minimally invasive double valve replacement surgery, the visual analog scale score for the thoracoscopic group remained above 3 on the third postoperative day, indicating moderate pain.⁹ After thoracic surgery, patients often experience moderate to severe pain, which, if not properly managed, can progress to chronic pain. In a randomized controlled trial (RCT) of minimally invasive direct coronary artery bypass grafting, patients who used patient-controlled intravenous analgesia alone postoperatively had an inspiration numeric rating scale score of 5.3 during the first 6 hours, which could negatively affect sleep quality and respiratory rehabilitation.¹⁰ Postoperative pain not only causes discomfort and extends hospitalization, but may also lead to pulmonary complications, pneumonia, and even chronic postoperative pain syndrome.¹¹ A prospective cohort study indicates that individuals undergoing minimally invasive aortic valve surgery are expected to have their pain, anxiety, and depression scores revert to baseline levels 1 month after surgery.¹²

Despite advances, both acute and chronic postoperative pain continue to present substantial challenges,¹³ highlighting the necessity for effective perioperative pain management. The question of whether MICS results in postoperative pain continues to be a topic of discussion among researchers. In

traditional cardiac surgery, pain management typically involves large doses of opioids to mitigate physiological responses to surgical pain and provides postoperative patient-controlled intravenous analgesia and salvage analgesia. For some patients with severe postoperative pain, even continuous infusion of remifentanyl may be necessary. However, prolonged opioid use has been associated with increased risks of pneumonia, extended intubation times, and higher mortality rates. Furthermore, opioid use has been linked to longer stays in intensive care units (ICUs) and extended hospitalizations following cardiac surgery.¹ As a result, the Enhanced Recovery After Surgery program is gaining popularity, emphasizing the need to explore more effective perioperative analgesic regimens for minimally invasive cardiac surgeries. These strategies include multimodal analgesia, opioid-reduced anesthesia, and long-term fascia plane blocks.^{1,14}

In the context of minimally invasive surgery, regional anesthesia has emerged as a key focus for cardiac anesthesiologists, particularly due to the smaller incision sites involved.¹⁵ A multimodal approach not only reduces the dependence on opioid analgesia but also enhances perioperative pain management. This approach proves to be an effective element in managing moderate pain throughout the perioperative phase. However, further research is necessary to fully assess whether various nerve blocks can deliver high-quality analgesia, decrease opioid consumption, alleviate postoperative discomfort, and prevent the development of chronic postoperative pain. The aim is to review the application of different nerve blocks in MICS, emphasizing their unique characteristics, comparing the advantages and disadvantages of each technique, and suggesting directions for future clinical research and comparative studies.

Neural Blockade Methods for MICS

Neural blockade plays an essential role in managing pain during MICS. Thoracic epidural anesthesia (TEA) and thoracic paravertebral plane block (TPVB) remain the most commonly used techniques, regarded as the “gold standards” for thoracic regional analgesia in clinical practice.² In recent years, alternative methods such as spinal anesthesia (SPA) and fascial plane blocks have gained considerable attention. Techniques like erector spinae plane block (ESPB), serratus anterior plane block (SAPB), pectoral nerve block (Pecs block), ICBN, and parasternal block (PSB) are increasingly being explored as viable alternatives.¹⁶

A summary of the advantages and disadvantages of these regional anesthesia techniques is provided in Table 1.^{17–26} The choice of a regional analgesic method is determined by various factors, including the nature of the surgical procedure, the individual requirements of the patient, and the proficiency of the anesthesiologist.¹³ Regional anesthesia may be applied either before or after the surgical procedure. Compared with postoperative regional anesthesia, the goal of preoperative regional anesthesia is to minimize opioid use during the procedure and make intraoperative hemodynamics more stable. However, the comparative effectiveness of preoperative versus postoperative

Table 1
Summary of the Regional Techniques^{17–26}

Regional Anesthesia	Nerves Affected	When to Use	Advantages	Disadvantages
TEA	Spinal nerve root	In all kinds of minimally invasive cardiac surgical procedures	The most effective analgesic, gold standard Early extubation Decreased pulmonary complications Reduce myocardial ischemia and improve left ventricular function	Hematoma risk Low blood pressure risk
SPA	Directly acting on the central nervous system	In all kinds of minimally invasive cardiac surgical procedures	Reduce the use of opioid drugs Improve respiratory function Compared with TEA, the operation is simpler Lower hematoma risk than TEA	Research results in heart surgery are limited Hematoma risk
TPVB	Spinal nerve	In all kinds of minimally invasive cardiac surgical procedures	Effective analgesic Lower hematoma risk than TEA	Low blood pressure risk
SAPB	3rd–6th intercostal nerves, the long thoracic nerve and the thoracodorsal nerve	Surgical incision on the lateral chest wall	Lower risk of hypotension Easy to perform and shorter the block application time Effective analgesic Less opioid consumption than ESPB	Research results in heart surgery are limited The evidence of complications is limited Limited in hemisternotomy
ESPB	Dorsal and ventral rami of the thoracic spinal nerves and sympathetic fibers	Surgical incision on the lateral chest wall	Easier to preformed than TEA Has lower visual analog scale score than SAPB Lower opioid consumption than SAPB Postoperative respiratory function is better than patients with SAPB Provide shorter staying time Better quality of recovery	Need more studies and large amount of evidence Effectiveness of postoperative pain control in cardiac surgery is not well studied Limited in hemisternotomy
ICBN	Lateral cutaneous branch	Surgical incision on the lateral chest wall	Effective analgesic Have higher FEV ₁ , better pulmonary function, and less pulmonary complications in thoracotomy patients Less demand for rescue analgesics and shorter length of ICU stay for minimally invasive direct coronary artery bypass grafting patients Can be performed under direct surgical vision	To get adequate blockade, 5–6 single shots are needed Single shot of ICNB cannot last a long time Need more research and large amount of evidence in cardiac surgery Limited in hemisternotomy
Pecs I and Pecs II	Pectoral nerves, 3rd–6th intercostal nerves, intercostobrachial nerves and the long thoracic nerve	Surgical incision on the lateral chest wall	Easy to perform Decreased opioid consumption	Limited research Limited in hemisternotomy
PSB	Anterior branch of intercostal nerve	Hemisternotomy	Easy to operate Can be performed under direct surgical vision	Limited research Risk of internal mammary artery injury

ESPB, erector spinae plane block; FEV₁, forced expiratory volume in the first second; ICNB, intercostal nerve block; PSB, parasternal block; ICU, intensive care unit; SAPB, serratus anterior plane block; SPA, spinal anesthesia; TEA, thoracic epidural anesthesia; TVPB, thoracic paravertebral plane block.

regional anesthesia, as well as its overall impact on clinical outcomes, remains an area for further investigation.

Thoracic Epidural Anesthesia

Epidural anesthesia was first introduced in 1921 and has since become a widely accepted method in noncardiac surgeries, enhancing analgesic outcomes. In the context of chest surgery, TEA is considered the gold standard for postoperative analgesia.² A recent review of studies from 2018 to 2023 reinforces the continued efficacy of epidural blockade in providing anesthesia and managing acute pain, even after minimally invasive surgery and various fascial plane blocks.²⁷ Additionally, TEA has been shown to decrease the incidence of pulmonary complications.²⁸ A retrospective study revealed that, when performed in experienced centers, epidural analgesia is a safe technique for noncardiac surgeries, with a low complication rate, including minimal risks for hematoma, abscess, or permanent nerve damage.²⁹

Introduced in cardiac surgery in the 1990s,³⁰ TEA typically involves puncturing the T3-T4 or T4-T5 spinous process space, through which an epidural catheter is inserted to administer local anesthetics continuously. This method blocks the T1-T5 spinal nerves and sympathetic chain.³¹ TEA can be performed with the patient either awake or sedated, and it can be implemented preoperatively or postoperatively. It is suitable for a wide range of minimally invasive cardiac surgeries, particularly those involving midline incisions, such as hemisternotomy.

TEA provides significant pain relief, improves postoperative respiratory function, and reduces extubation times, particularly in elderly patients undergoing cardiac surgery.³² It also promotes coronary artery dilation, slows the heart rate, enhances myocardial metabolism, and improves coronary perfusion.³³ Studies have demonstrated that TEA can reduce myocardial ischemia and improve left ventricular function.³⁴ In pediatric cardiac surgery, thoracic epidural analgesia has been shown to lower perioperative blood pressure, improve blood glucose control, decrease inflammatory markers, and reduce the incidence of postoperative acute kidney injury.³⁵

Despite its status as the gold standard, cardiac anesthesiologists often hesitate to use TEA due to concerns over epidural hematoma, particularly because cardiac surgery patients are often on anticoagulant or antiplatelet therapy. The advantages of epidural anesthesia should be assessed meticulously in relation to the potential hazards of epidural hematoma. A systematic review indicates that the risk of developing an epidural hematoma during a cardiac surgery is roughly 1 in 7,643.³⁶ Although recent clinical trials have not reported any cases of epidural hematoma, isolated case reports persist, underscoring the importance of a comprehensive preoperative assessment of the patient's coagulation profile and anatomy. Postoperative neurological monitoring is also essential, particularly after the insertion or removal of epidural catheters in cardiovascular surgery.³² Another challenge is the need for high levels of technical proficiency, because incorrect block placement can

affect bilateral sympathetic nerves, potentially leading to hypotension. An RCT involving 1,802 patients revealed that the use of combined general and epidural anesthesia correlated with an increased occurrence of hypotension.³⁷ Furthermore, recent research indicates that epidural analgesia might extend the QT interval, probably as a result of sympathetic nerve blockade.³⁸

At present, research on TEA in MICS is limited, which may be attributed to its relatively low usage. Nevertheless, we anticipate that its application will continue to grow. Future studies should focus on reaffirming the benefits of epidural anesthesia in this setting and further clarifying its role in improving patient outcomes in MICS.

Spinal Anesthesia

SPA is widely used for the management of postoperative pain.³⁹ Intrathecal opioid administration, with or without local anesthetics, is commonly performed under SPA and proves highly effective in relieving postoperative discomfort.⁴⁰ This technique allows opioid drugs to rapidly reach the cerebrospinal fluid, enabling swift action on the central nervous system. Intrathecal morphine is commonly used in various surgical procedures, including thoracic, abdominal, spinal, urological, and orthopedic surgeries, and is considered the benchmark for analgesia during cesarean sections.⁴¹

High SPA has been used for decades in cardiac surgery patients.⁴² Clinical studies suggest that the puncture is typically performed before anesthesia induction, with the injection site often chosen at the L3-L4 or L4-L5 intervertebral spaces. The patient may be positioned sitting or laterally, with the plane adjusted to T1 using the Trendelenburg position.^{25,43} For MICS, the procedural approach is similar, with no significant differences in surgical techniques.

Currently, there is limited research on the postoperative analgesic effects of SPA in cardiac surgery, particularly in comparison with thoracic epidural analgesia. A 2021 RCT by Dhawan et al⁴⁴ demonstrated that intrathecal morphine significantly alleviated pain during rest and coughing for up to 48 hours after thoracoscopic bypass surgery, reducing opioid consumption. A recent meta-analysis corroborated these findings, showing that intrathecal morphine significantly decreased morphine use 24 hours after surgery compared with control groups in cardiac surgery patients.⁴⁵

The main complication associated with SPA is respiratory depression, necessitating close monitoring of respiratory function before extubation. However, studies indicate that intrathecal injection may actually reduce the time to postoperative extubation in cardiac surgery without causing respiratory depression.²⁶ Although many minimally invasive cardiac surgeries require single lung ventilation, limited clinical data exist on the effects of intrathecal injection on respiratory recovery in these procedures. Another concern is the potential for spinal hematoma. The 2018 American Heart Association guidelines recommend administering intravenous heparin 1 hour after neuraxial procedures, although specific evidence related to cardiac and minimally invasive cardiac surgeries is still limited.⁴⁶

Given the scarcity of research on SPA in MICS, further studies are essential to assess its efficacy and potential complications in this context.

Thoracic Paravertebral Block

The TPVB serves as a proficient method of regional anesthesia, offering substantial pain alleviation across various surgical interventions, such as those involving the breast and lungs.⁴⁷ In noncardiac surgeries, the paraspinal nerve block has a lower failure rate compared with TEA and can achieve comparable analgesic effects. TPVB also offers postoperative analgesia of similar quality to TEA while helping to stabilize hemodynamics.⁴⁸

In MICS, TPVB can be administered unilaterally or bilaterally. It is particularly beneficial for surgeries involving midline incisions, such as sternotomy, with bilateral blockade required for hemisternotomy. Bilateral sympathetic blockade theoretically improves myocardial oxygen balance.⁴⁹ TPVB can be performed both preoperatively and postoperatively. The procedure involves injecting local anesthetics into the paraspinal space to block spinal nerve branches. This space, located between the thoracic spinal nerve and the intervertebral foramen, lacks fascia, allowing small amounts of anesthetic to effectively alleviate pain and influence the sympathetic nervous system. Adjacent paraspinal spaces facilitate anesthetic diffusion. TPVB can effectively anesthetize 1 to 3 segments on the same side as the injection, similar to unilateral epidural analgesia.⁵⁰ Ultrasound guidance has enhanced the safety and precision of this technique, with several methods reported.⁵¹ Continuous paraspinal block has also been shown to provide longer-lasting pain relief.⁵²

A meta-analysis indicates that continuous TPVB provides analgesia comparable to epidural anesthesia in cardiothoracic surgeries.⁵² Recent cohort studies suggest that preoperative bilateral TPVB facilitates early extubation and reduces intraoperative and postoperative fentanyl consumption in cardiac surgeries.⁵³ Studies have shown that TPVB's analgesic efficacy after MICS is similar to that of TEA.

For most MICS, TPVB can be performed unilaterally, preserving sympathetic function on the opposite side.⁵⁴ This factor contributes to a lower incidence of hypotension compared with TEA. Moreover, TPVB is associated with fewer incidents of urinary retention, nausea, and vomiting compared with TEA.⁵⁵ Despite the reduced risk of epidural hematoma, careful monitoring for hypotension remains crucial.⁵⁶

Based on the available literature from the past five years, clinical trials on the application of TPVB in MICS are relatively limited. However, the technique provides effective pain relief, and the catheterization site is far from the incision, making it particularly suitable for post-surgical use. Despite this, further research is needed to investigate the effects of heparinization on hematoma formation and TPVB in cardiac surgery, as well as other potential complications. Higher-quality studies are essential to confirm its safety and efficacy.

Erector Spinae Plane Block

The ESPB has gained recognition as an effective postoperative analgesic approach in both thoracic and cardiac surgeries.⁵⁷ In thoracic procedures, studies have demonstrated that ESPB significantly reduces opioid prescriptions,⁵⁸ and lowers numeric rating scale scores.⁵⁹ Additionally, no notable differences have been observed in the duration of ventilator use or ICU stay between ESPB and TEA.⁶⁰

This method entails the administration of local anesthetics into the erector spinae muscle and adjacent tissues, with accurate positioning facilitated by ultrasound guidance.⁶¹ In the context of thoracic and cardiac surgical procedures, the administration of injections is generally performed at the T4 or T5 vertebral levels. This maneuver can be executed as a singular injection or through the placement of a catheter, with the objective of obstructing both the dorsal and ventral rami of the thoracic spinal nerves, in addition to the sympathetic fibers.^{16,62} A meta-analysis indicated that the application of single-needle ESPB may enhance clinical outcomes for patients undergoing cardiac surgery through sternotomy, although the enhancements in pain scores, recovery of respiratory function, and duration of hospital stay were relatively modest.⁶³ In studies involving cadavers, the dye used for erector spinae muscle plane blockade has demonstrated diffusion to the anterior chest wall.⁶⁴ However, additional clinical trials are necessary to verify the degree of sensory loss in this area.

Recent research has explored the application of ESPB in cardiac surgery and minimally invasive cardiac interventions. Observations suggest that ESPB is more straightforward to execute compared with TEA.⁶⁰ A comprehensive review indicated that ESPB led to a reduction in pain scores within 12 hours after extubation in cesarean section patients and resulted in decreased opioid use 24 hours after surgery.⁶⁵ Another meta-analysis, however, indicated that, in patients undergoing sternotomy, the ESPB did not lead to a significant reduction in postoperative pain scores or opioid consumption. This finding suggests that the efficacy of this particular intervention may be constrained in individuals subjected to midline incision surgeries.⁶⁵ In MICS, the use of the erector spinae plane catheter has shown promise in improving postoperative outcomes, providing better pain relief and reducing ICU and hospital lengths of stay.⁶⁶ Its analgesic effectiveness has been found to surpass that of the SAPB.⁶⁷ However, there remains ongoing debate regarding opioid consumption, based on findings from different research centers,^{68–70} and further RCT are needed to validate these conclusions. The potential link between ESPB and enhanced recovery remains an area of active research.⁷¹

Although ESPB is generally regarded as simpler and safer than TEA and TPVB, substantial evidence supporting its efficacy remains insufficient. A case report has suggested that ESPB may provide broader skin coverage and fewer complications compared with TEA.⁶⁰ However, more research on clinical concerns, such as bleeding risks, is necessary to fully assess its safety and effectiveness.

Serratus Anterior Plane Block

The SAPB is commonly used in thoracic surgery, breast surgery, and for managing rib fractures. SAPB has demonstrated comparable performance to other analgesic techniques. In contrast with local infiltration anesthesia, preoperative SAPB administration results in lower visual analog scale scores and reduced use of rescue medications. However, its impact on intraoperative opioid consumption remains inconclusive.^{72,73} Several studies, including meta-analyses, have consistently shown that combining SAPB with patient-controlled analgesia significantly reduces opioid use.^{74,75} In terms of respiratory function, postoperative forced expiratory volume in the first second (FEV₁) and forced vital capacity were found to be superior in the SAPB group.¹⁹

Ultrasound-guided SAPB was first introduced in 2013⁷⁶ and is performed at the midaxillary line at the fifth rib. The technique involves injecting local anesthetics into the surface or deeper layers of the serratus anterior muscle to block the 3rd to 6th intercostal nerves, the thoracic long nerves, and the thoracic dorsal nerves.⁷⁷ Although there are two methods for performing SAPB, studies have shown no difference between them in terms of remifentanyl consumption, emergence time, rescue medication use, or variations in blood pressure and heart rate at different time points.⁷⁸

Research on SAPB in cardiac surgery remains limited. A prospective observational cohort study in Italy found that continuous SAPB effectively reduced opioid demand in patients undergoing minimally invasive mitral valve surgery, but there was no significant difference in ICU or hospitalization time.⁷⁹ Although evidence regarding complications remains sparse, SAPB is considered a safe and effective analgesic technique in thoracotomy in cardiac surgery.⁸⁰ SAPB also holds promise as a valuable option in MICS.

Because SAPB does not inhibit sympathetic nerve activity, it reduces the risk of hypotension, making it suitable for patients on anticoagulant therapy. As a result, SAPB offers a viable alternative for patients who are not suitable candidates for TPVB or TEA. Zhang et al's¹⁷ RCT demonstrated that, although DSPB and SSPB provide similar analgesic effects, they are easier to perform than TPVB. Baytar et al's⁸¹ research suggests that SAPB reduces the duration of block application, positioning it as a feasible alternative to TPVB.

Pecs Block

Chest wall fascia blocks were first introduced in breast surgery and have since been used for postoperative analgesia in a variety of chest wall surgeries. However, research focusing on chest wall nerve blocks remains somewhat limited. Studies indicate that Pecs II reduces pain scores and opioid consumption within 24 hours after minimally invasive thoracic repair surgery.⁸²

Pecs blocks target the pectoral nerves, including the 3rd to 6th intercostal nerves, intercostobrachial nerves, and long thoracic nerves. Two fundamental techniques exist: Pecs I and Pecs II. Both methods exhibit simplicity and ease of

administration. Pecs I, which was introduced in 2011, features a high-capacity fascial block situated between the pectoralis major and pectoralis minor muscles. In contrast, Pecs II, introduced in 2012, focuses on the 3rd to 6th intercostal nerves, intercostobrachial nerves, and long thoracic nerves by obstructing the area between the pectoralis minor and the serratus anterior muscles.⁸³ Both techniques are simple to perform and can be applied either preoperatively or postoperatively.

Pecs blocks also provide benefits in MICS, including pain relief, reduced opioid consumption, and shorter hospital stays.²¹ A retrospective study confirmed that Pecs II is a safe and effective option for regional analgesia in robotic mitral valve surgery, with efficacy.⁸⁴

In theory, due to the injection site, the likelihood of hematoma and hypotension is lower compared with TEA as an analgesic method. However, research in cardiac surgery is currently very limited and more clinical studies are needed.

Intercostobrachial Nerve Block

ICNB is widely used for perioperative analgesia and pain management in a range of thoracic and abdominal surgeries, showing clear and consistent results. Research has shown that ICNB decreases the need for opioids⁸⁵ and enhances lung ventilation in individuals with sternal fractures and those undergoing thoracotomies.^{86,87}

In ICNB, 3 to 5 mL of local anesthetic are injected near the intercostal nerves, often requiring 5 to 6 injections for adequate blockade.⁶² Under ultrasound guidance, the needle is positioned at the inner layer of the intercostal muscles, with anesthetic injected both above and below the incision.⁸⁸ ICNB can be administered at the start or end of surgery, serving as an effective pain control measure. Although the effect of a single ICNB injection is relatively short lived, continuous injections offer a longer duration of action. Ma's⁸⁹ study found no significant difference in pain control between patients receiving continuous ICNB and those receiving continuous TPVB. Compared with patients who received only PVB, those undergoing intercostal cryoanalgesia experienced improved pain control,⁹⁰ with postoperative FEV₁ almost returning to preoperative levels.⁹¹ Some studies suggest that ICNB can be an alternative to TEA, providing similar pain relief following minimally invasive surgery up to the 5th postoperative day.^{92,93}

In cardiac surgery, some studies report that continuous ICNB can effectively reduce postoperative pain in patients undergoing minimally invasive mitral valve repair surgery.^{94,95} This finding is particularly relevant for posterolateral thoracotomy, where intercostal nerve fibers may be severely compromised by the use of metal rib retractors.⁹⁶ Research supports that ICNB can prevent post-thoracotomy pain, with protection of the intercostal nerves thought to contribute to alleviating postoperative discomfort.⁹⁷ A retrospective cohort study in China suggested that continuous ICNB provides significant benefits for patients undergoing minimally invasive direct coronary artery bypass grafting, including a reduction in the need for rescue analgesics and shorter ICU

stays.⁹⁸ Additionally, a study involving pediatric cardiac surgery patients found that ICNB can reduce extubation time.⁹⁹ Furthermore, the novel technique of intercostal cryo nerve block in minimally invasive thoracotomy heart valve surgery has been shown to improve FEV₁ scores 2 days after surgery.¹⁰⁰

However, the analgesic effect of a single ICNB is relatively limited in scope, often requiring the blockade of multiple intercostal nerves to achieve adequate pain relief. Additionally, incomplete parietal pleura may hinder the local accumulation of drugs. Currently, there is no evidence suggesting that ICNB offers neuroprotective effects, though this may become an area for future exploration.

Parasternal Blocks

PSB have seen increasing application in sternotomy surgeries in recent years, with the majority of research focusing on cardiac surgery. However, relevant studies also exist in breast surgery and the implantation of implantable cardioverter-defibrillators.^{22,101}

The parasternal block specifically focuses on the anterior cutaneous branch of the intercostal nerve.¹⁰² Parasternal intercostal plane blocks can be classified into two distinct categories: deep and superficial. In both variations, the anesthesiologist must place the probe near the sternum, using color Doppler to gradually shift the probe outward to locate the internal thoracic artery and examine the intercostal space between the fourth and fifth ribs as the artery becomes discernible. In the deep parasternal intercostal plane block, the injection occurs in the space between the intercostal muscles and the transverse pectoralis muscle.¹⁰³ The injection for the superficial parasternal intercostal plane block is administered in the space between the pectoralis major and the intercostal muscles.¹⁰⁴ Using an in-plane approach, the needle is positioned within the fascial plane that exists between the pectoralis major muscle and the pleura.¹⁰⁵ This method may also be used for localized anesthesia under direct observation before the closure of the sternum. MICS demonstrates a notable suitability for hemisternotomy procedures.

In cardiac surgery, a recent meta-analysis suggests that the superficial parasternal block, administered through sternotomy, may improve acute postoperative pain management.¹⁰⁶ Research by Chen indicates that bilateral sustained superficial parasternal intercostal plane block blockade reduces pain scores both at rest¹⁰⁷ and during activity,¹⁰⁸ while also decreasing opioid consumption within 48 hours.¹⁰⁸ Additionally, studies have demonstrated that the deep parasternal intercostal plane block group exhibits more stable hemodynamic parameters during sternotomy,¹⁰⁹ reduced intraoperative opioid use,¹¹⁰ and shorter extubation times with lower rates of postoperative nausea.¹¹¹ Case reports have also suggested that this technique may alleviate chronic pain after surgery.²³ However, additional high-quality research is needed. Current studies primarily focus on patients undergoing median incision cardiac bypass surgery, with relatively limited research in MICS.

No complications associated with this technique have been reported to date. In cardiac surgery involving a median

incision, concerns include the risk of bleeding and potential damage to the internal mammary artery.¹¹² Research on the use of this technique in MICS, particularly with hemisternotomy, is currently lacking. Considering the challenges associated with targeting the anterior cutaneous branches of intercostal nerves through lateral chest wall blocks, methods like TEA and TPVB present inherent risks, such as epidural hematoma and hypotension. Consequently, the parasternal block could present a more secure and efficient option for postoperative pain management after hemisternotomy; however, additional high-quality evidence is required.

Opioids and Local Anesthetics

Intrathecal drugs include local anesthetics, such as bupivacaine, and opioids, such as morphine and sufentanil, with morphine being the most commonly used. Studies have shown that low to moderate doses of intrathecal morphine, ranging from less than 0.3 mg to 0.3 to 0.5 mg, can provide analgesic effects while reducing complications in patients undergoing cardiac surgery.¹¹³ However, research on intrathecal injection in MICS is limited. Morphine remains the preferred drug for intrathecal administration, with studies suggesting a dose of 5 µg/kg provides adequate analgesia while minimizing adverse reactions in MICS.⁴⁴ Further clinical trials are needed to determine the optimal dosage.

For regional nerve blocks, bupivacaine and ropivacaine are the most frequently used local anesthetics. In MICS, bupivacaine concentrations typically range from 0.125% to 0.500%, and ropivacaine is used at concentrations between 0.2% and 0.5%, depending on the blocking method, injection site, and volume. To achieve a broader blockade, researchers often use the maximum allowable dose based on the patient's tolerance.

To prolong the blocking effect, most studies use catheterization; however, the studies reviewed here did not mention the addition of adjuvants. Previous noncardiac surgical studies have demonstrated that hydrocortisone and methylprednisolone can extend the duration of blockade.^{114,115} Furthermore, both dexmedetomidine and nalbuphine have been identified as adjuvants for anterior serratus muscle plane blocks, which can contribute to maintaining hemodynamic stability.^{116,117} Although opioids are not typically used as adjuvants for peripheral nerve blocks, research on other types of blocks remains limited.

Bupivacaine liposomes, a newer long-acting local anesthetic, have a recommended maximum dose of 266 mg. There are reports on its use in ICNBs, SAPBs, and Pecs blocks,^{118,119} but no relevant studies exist on other types of blocks. Further research is required to assess the action duration, dosage, safety, and efficacy of bupivacaine liposomes for postoperative analgesia in MICS.

To achieve a more long-lasting analgesic effect, cardiac anesthesiologists choose to insert catheters. However, in current related studies, local anesthetics are injected through micro pumps. Even in the research related to thoracic epidural anesthesia, there are very limited reports on patient-controlled epidural analgesia. The implementation of patient-controlled

epidural analgesia in cardiac anesthesia requires more clinical practice and research.

Conclusions

Regional anesthesia in minimally invasive thoracic surgery has been shown to effectively reduce pain and promote recovery, as outlined in Enhanced Recovery After Surgery guidelines.¹²⁰ Techniques such as TEA and TPVB have proven to be consistently effective. However, in cardiac surgery, concerns regarding the risk of spinal hematoma with TEA persist, and research on its application in MICS is still limited. The regional blocks discussed in this paper may offer safe alternatives, with some potentially reducing opioid consumption. However, it remains uncertain whether these techniques can lower complication rates or enhance recovery outcomes following MICS. Currently, SPA stands out as a promising approach, being applicable to all minimally invasive cardiac procedures. In theory, SPA provides analgesic effects similar to TEA but carries a lower risk of hematoma. Although existing research supports the effectiveness of SPA, it does not definitively address whether ESPB or SAPB provides superior pain relief with fewer complications. Moreover, studies on ICNB and Pecs block in cardiac surgery remain limited. Superficial parasternal intercostal plane block and deep parasternal intercostal plane block are novel techniques that focus on mid-line incision pain management. Although these methods show potential, they are still under investigation in the context of MICS, and research remains sparse. Current studies mainly focus on case reports, cohort studies, and single center RCTs. Multiple center RCTs and meta-analysis reports are relatively limited. Further large-scale, prospective studies are necessary to assess the safety and efficacy of these techniques.

Declaration of competing interest

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CRediT authorship contribution statement

Kexin Yuan: Writing – original draft, Methodology, Investigation, Conceptualization. **Boqun Cui:** Writing – review & editing. **Duomao Lin:** Writing – review & editing. **Haiyan Sun:** Writing – review & editing, Funding acquisition. **Jun Ma:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

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